Nearest Neighbor Reasoning

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Recall and choose your starting point

> Record your starting explanation for Nearest Neighbor Reasoning before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Similarity by Selected Traits

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sampling and Representation

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Overfitting Intuition

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Nearest Neighbor Reasoning trace and record the first step you can explain without help.

Use W2: Apply **neighbour** and **tie rule**s for Nearest **neighbour**s and **distance**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Nearest Neighbor Reasoning explanation against the supplied facts before discussing it.

Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: “A negative difference is a negative **distance**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Find neighbours independently

New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3.

Using the same **training** pool and k=3, score validation (6,blocked),(10,clear),(12,blocked). Compare the **training**-majority baseline.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Nearest Neighbor Reasoning target to a trace and a justified conclusion.

A query has nearest books A and B at **distance** 1, with conflicting labels; C is **distance** 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Nearest Neighbor Reasoning case and show the deciding evidence.

A=[1,1] liked, B=[1,0] disliked, C=[0,0] disliked; query=[0,1].

Find nearest **neighbour**s using differing-trait **distance** and k=1.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Training items have one numeric feature: A=2 with label short, B=8 with label long. A new item has value 6. Use absolute **distance** to choose its nearest **neighbour**.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 22 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Nearest Neighbor Reasoning adds to Transparent Book Recommender and keep the evidence.

Use **neighbor**ing liked books as **evidence**, while explaining which traits created closeness.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A query has nearest books A and B at **distance** 1, with conflicting labels; C is **distance** 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Calculate feature distances

> Calculate feature distances. Apply the supplied Nearest Neighbor Reasoning facts and keep your reasoning.

For query 3 cm, calculate **distance**s to all five **training** rows. Explain why A and B tie even though their labels differ.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Apply neighbour and tie rules

> Apply neighbour and tie rules. Apply the supplied Nearest Neighbor Reasoning facts and keep your reasoning.

For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Keep evaluation examples separate

> Keep evaluation examples separate. Apply the supplied Nearest Neighbor Reasoning facts and keep your reasoning.

Why must the three validation reference labels not enter the **neighbour** pool? Give the fixed k=3 score and baseline score on the worked validation examples.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Find neighbours independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Find neighbours independently. Apply the supplied Nearest Neighbor Reasoning facts and keep your reasoning.

New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3.

Using the same **training** pool and k=3, score validation (6,blocked),(10,clear),(12,blocked). Compare the **training**-majority baseline.

My reasoning and evidence

W5 Interpret a vote and a match

> Interpret a vote and a match. Apply the supplied Nearest Neighbor Reasoning facts and keep your reasoning.

Two of three **neighbour**s **vote** clear. Does that prove a two-thirds chance of a correct **prediction**? Does an exact feature match guarantee a correct label? Explain both limits.

My reasoning and evidence

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Nearest Neighbor Reasoning

> A fictional one-feature classifier predicts compact or extended book format. Explain the labelled step before checking the changed condition.

A fictional one-feature classifier predicts compact or extended book format.

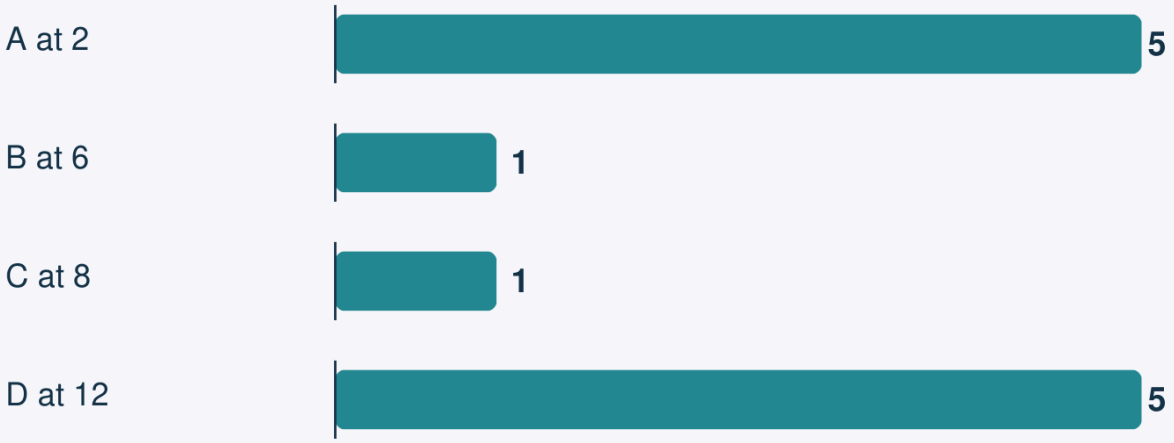
Stored **training** rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

Distance is absolute difference; ties sort by ID; tied category **vote** chooses compact.

AI ACADEMY / AI-2/22

Distances from query 7

absolute distance units



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/22

Nearest Neighbor Reasoning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Measure distances

From 7: A=5, B=1, C=1, D=5.



2. Order neighbours

Distance then ID gives B,C,A,D.



3. k=1 prediction

First neighbour B is extended: predict extended.



4. k=3 prediction

B,C,A vote extended,compact,compact: predict compact.

CHANGE ONE THING

Move query from 7 to 9 while keeping training fixed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why do k=1 and k=3 disagree for the original query?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional one-feature classifier predicts compact or extended book format.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B,C,A **vote** extended,compact,compact: predict compact.

> 1. k=1 **prediction**

2. Order **neighbour**s

3. k=3 **prediction**

4. Measure **distance**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: k=3 **prediction** B,C,A **vote** extended,compact,compact: predict compact. Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed.

The query's reference label must not enter the stored pool before predicting. Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move query from 7 to 9 while keeping **training** fixed. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended. **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Repair a missing building block

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing..

Return to this week and explain what you changed.

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can **80%** response be reported as **80%** of the whole grade responding?. Return to this week and explain what you changed.

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

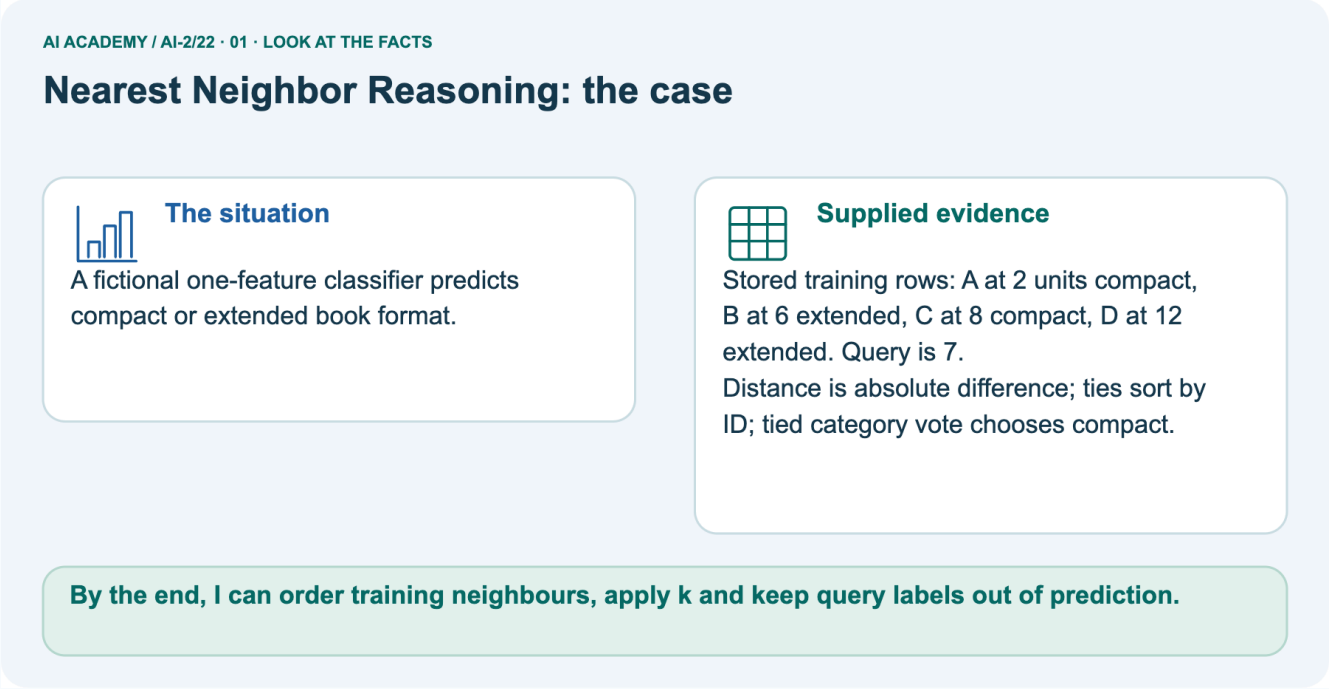
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Nearest Neighbor Reasoning picture from its labels, then record a relevant limit.

01. Nearest Neighbor Reasoning: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

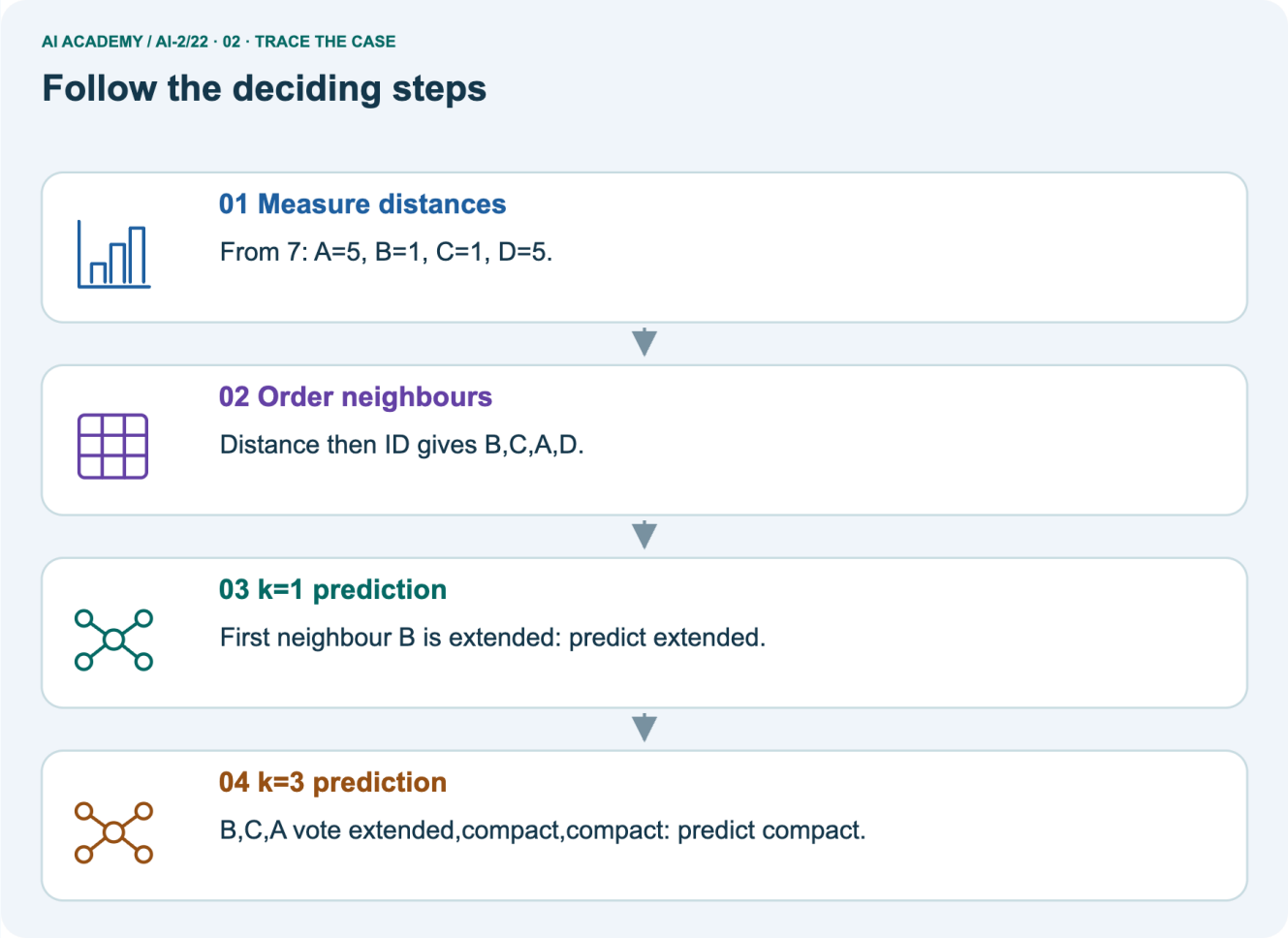
A fictional one-feature classifier predicts compact or extended book format. Supplied facts: Stored **training** rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

Distance is absolute difference; ties sort by ID; tied category **vote** chooses compact.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



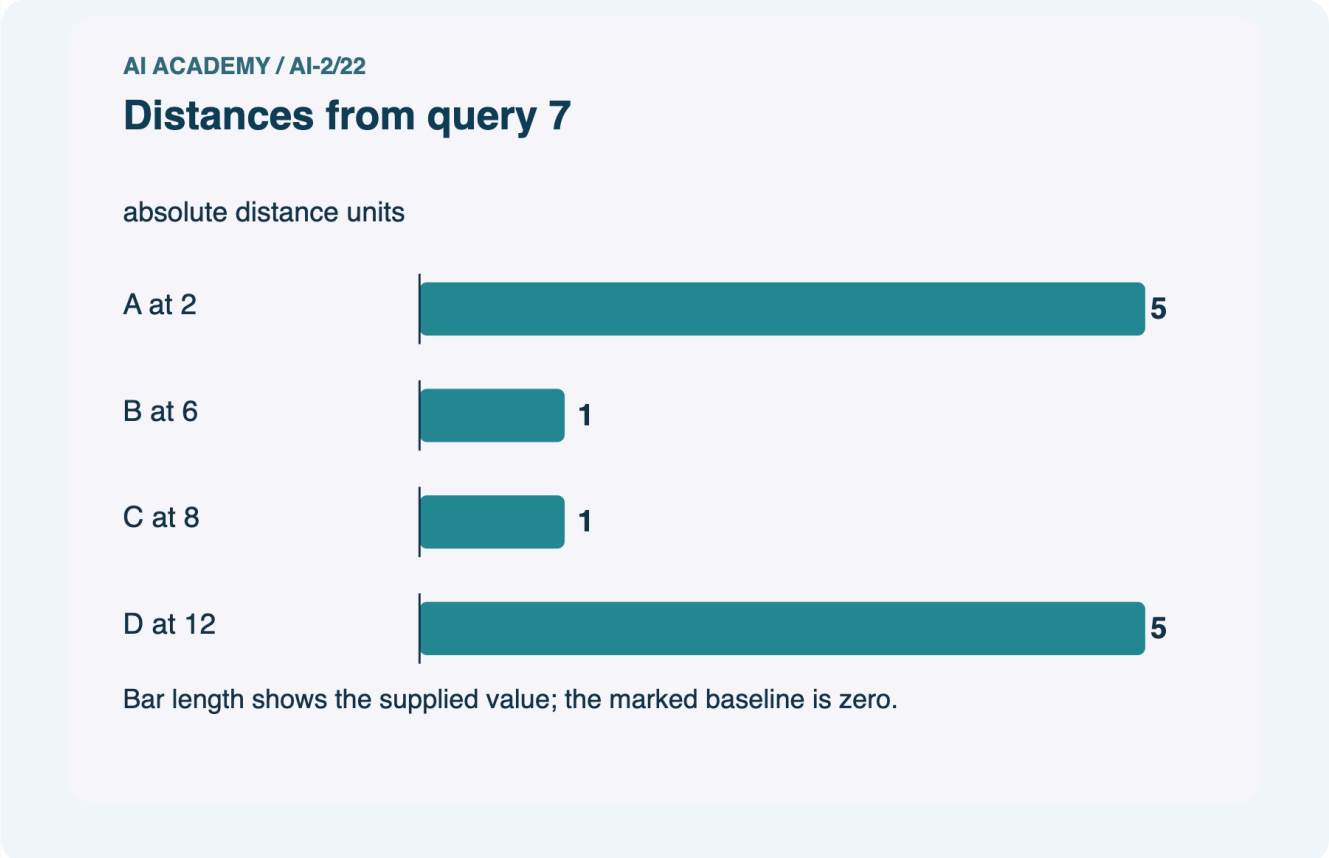
Follow the labelled picture with the matching explanation. Open the vector diagram

Measure **distance**s: From 7: A=5, B=1, C=1, D=5. → Order **neighbour**s: **Distance** then ID gives B,C,A,D. → k=1 **prediction**: First **neighbour** B is extended: predict extended. → k=3 **prediction**: B,C,A **vote** extended,compact,compact: predict compact.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Distances from query 7



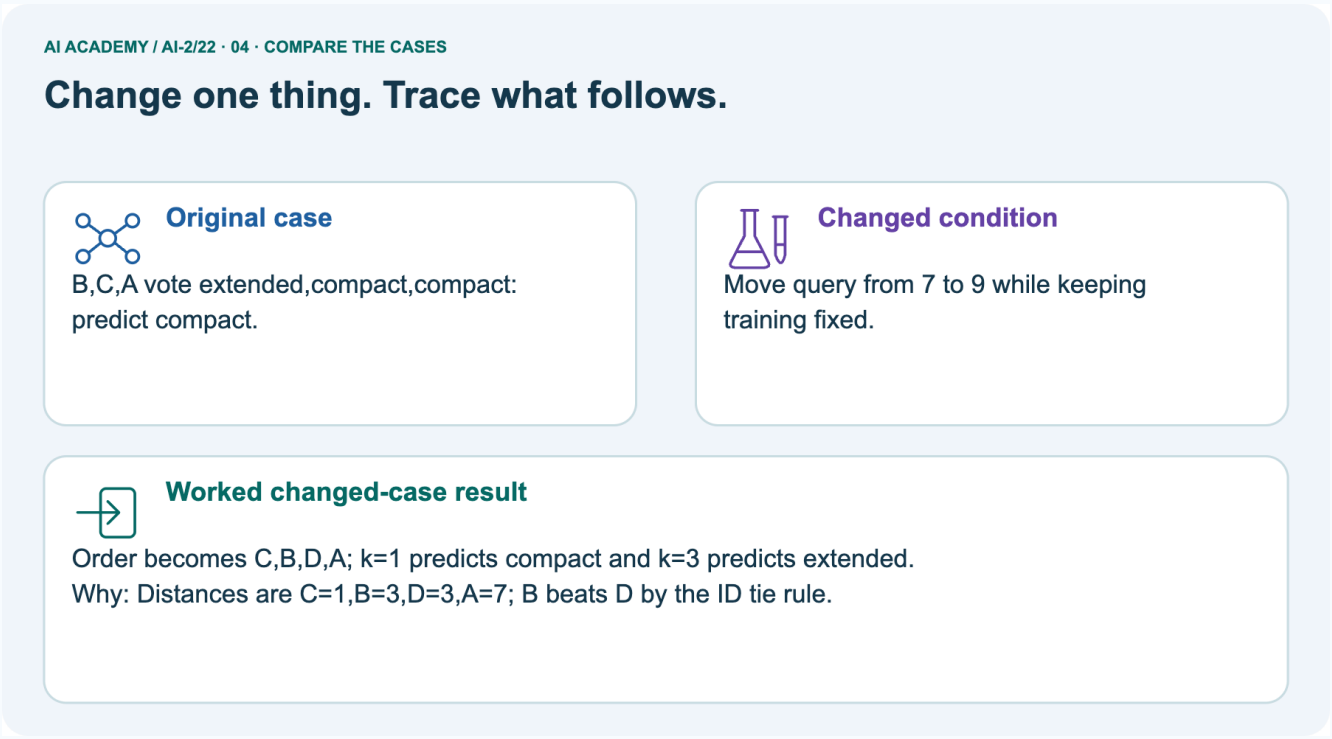
Follow the labelled picture with the matching explanation. Open the vector diagram

Distances from query 7. Units: absolute **distance** units. A at 2: 5. B at 6: 1. C at 8: 1. D at 12: 5.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

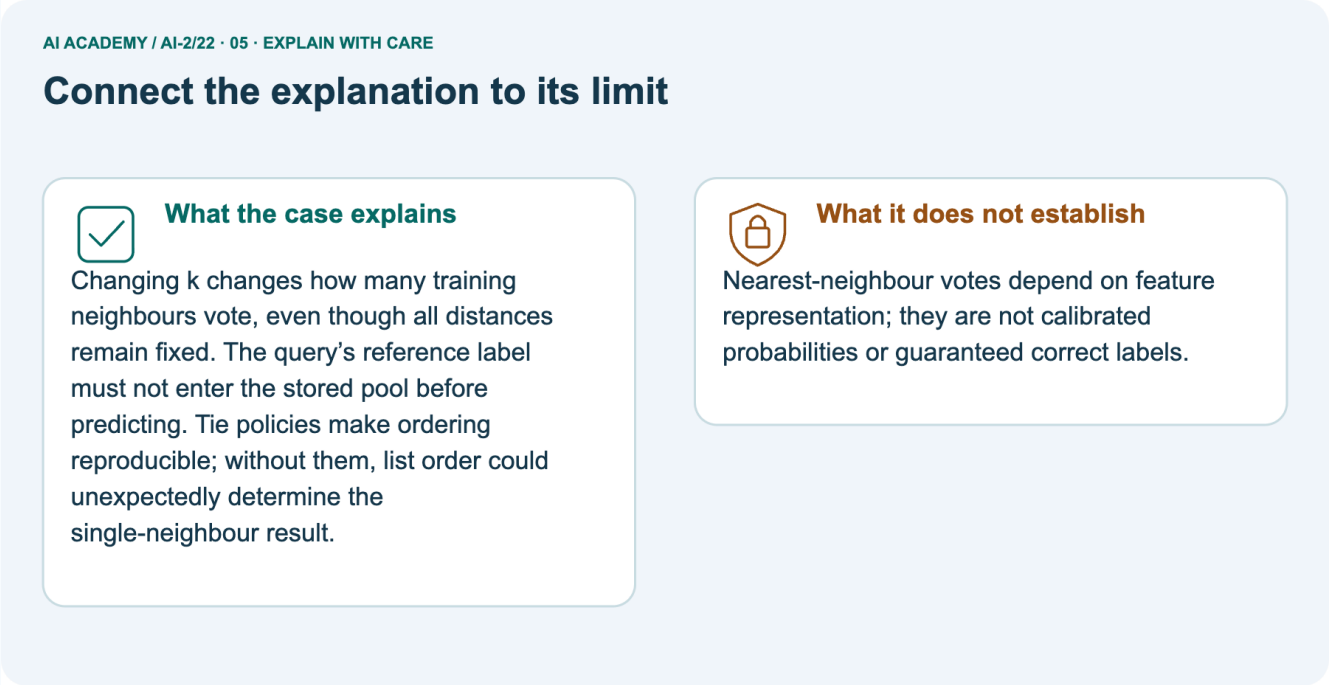
Original case: B,C,A **vote** extended,compact,compact: predict compact. Changed condition: Move query from 7 to 9 while keeping **training** fixed.

Expected result: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended. Reason: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed. The query's reference label must not enter the stored pool before predicting.

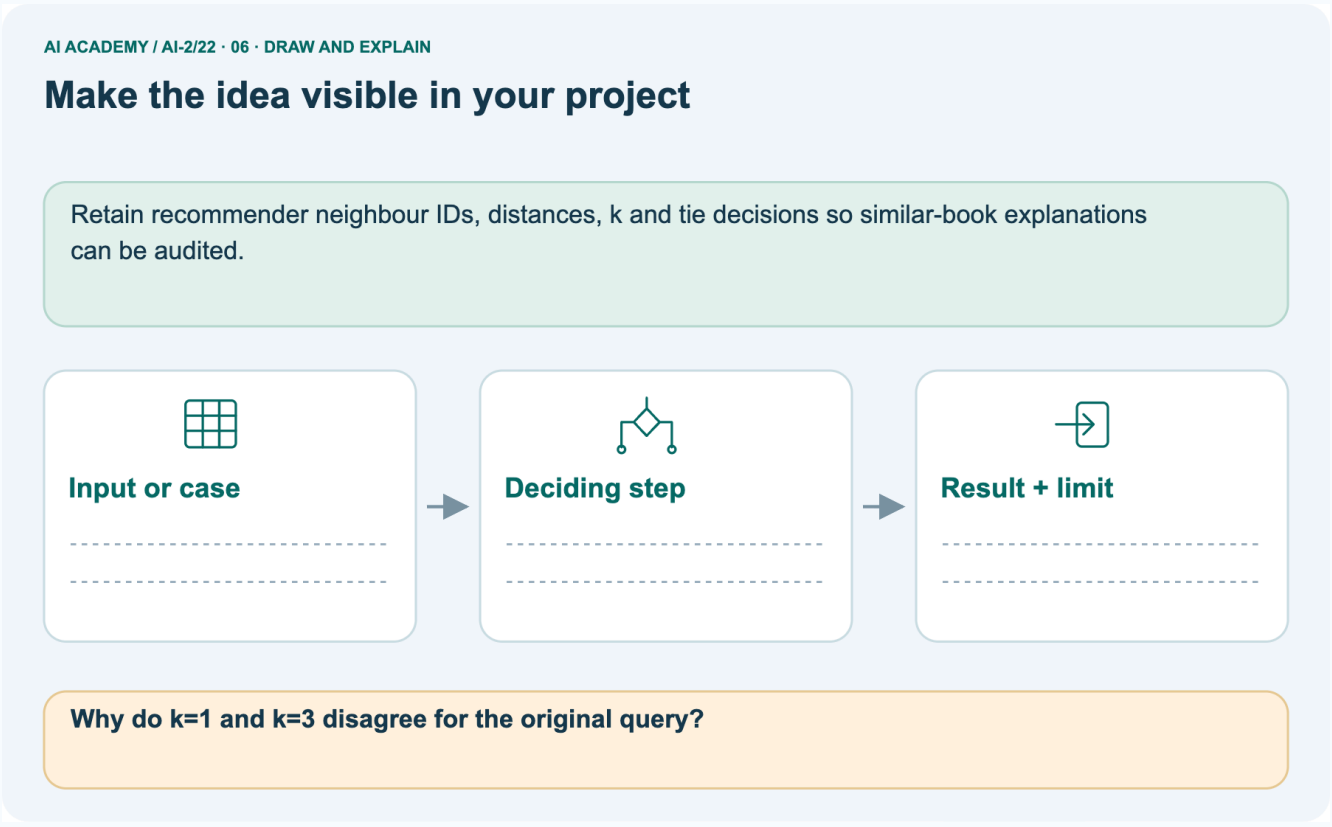
Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

Limit: Nearest-**neighbour vote**s depend on feature representation; they are not calibrated probabilities or guaranteed correct labels.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited. Draw your **input**, deciding step, result and limit.

Supported question: Why do k=1 and k=3 disagree for the original query?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.